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Retrieval-Augmented Generation

RAG





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Why do we need RAG?

RAG



The Problem with LLMs

RAG

Have you ever asked ChatGPT something it didn't know?

Company
policies?

Internal
documents?

Latest news?

Yesterday's
sales report?

LLMs cannot magically know information they were never trained on.





The Problem with LLMs

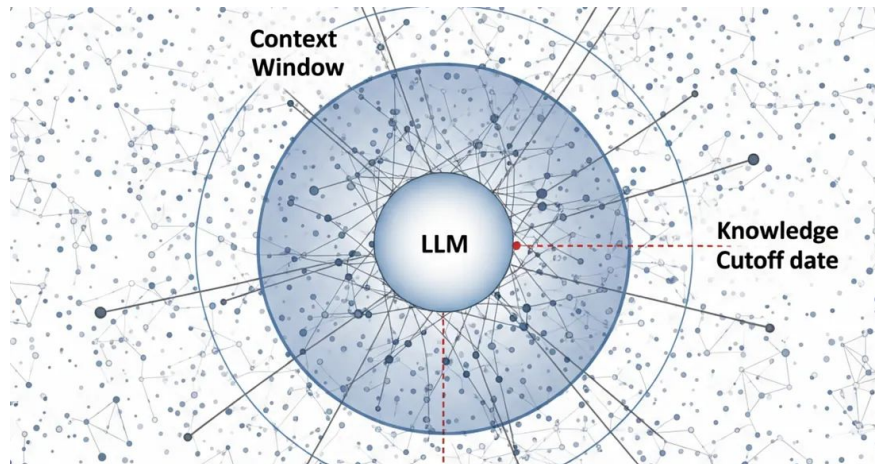
RAG

WHY?

LLMs learn from training data.

They **do not automatically know**

- New information
- Private company documents
- Your PDFs
- Internal knowledge bases



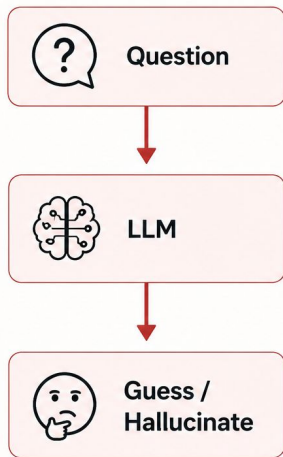
<https://sophiehundertmark.medium.com/llm-insight-why-gemini-notices-more-and-what-knowledge-cutoff-really-means-6cb44ba6ab09>



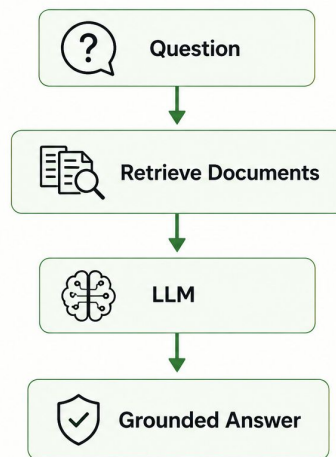
What problem does RAG solve?

RAG

Without RAG



With RAG





Why is RAG Important?

RAG

Uses your own
data

More Grounded
Answers

Accesses Updated
Information

No Model
Retraining



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What is RAG?

RAG



What is Retrieval-Augmented Generation?

RAG

Retrieval-Augmented Generation (RAG) is a technique that combines

Information Retrieval

X

Large Language Models

to generate more grounded answers.



<https://arbisoft.com/blogs/how-rag-pipelines-power-analytics-with-llms-and-data-engineering>



Why and When should we use RAG?

RAG

Why?

Instead of relying only on memory,
the LLM can
look up information first.

When?

Use RAG whenever information

- changes frequently
- is private
- is too large to memorize



Retrain
a billion-parameter
model

Retrieve
one relevant
document



RAG Analogy

RAG

Closed-book Exam

LLM without RAG



Closed-book



- No external resource
- Higher chance of hallucination

Open-Book Exam

LLM with RAG



Open-book



- Can look up relevant information
- More accurate and reliable answers



Limitations

RAG

RAG is powerful, but not perfect.

Limitations

- Poor retrieval → Poor answers
- Missing documents
- Incorrect chunks
- Slow retrieval on huge datasets
- Depends on embedding quality

RULE:

Better Retrieval → Better Answers

Retrieval quality strongly affects the generated response.



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How RAG Works?

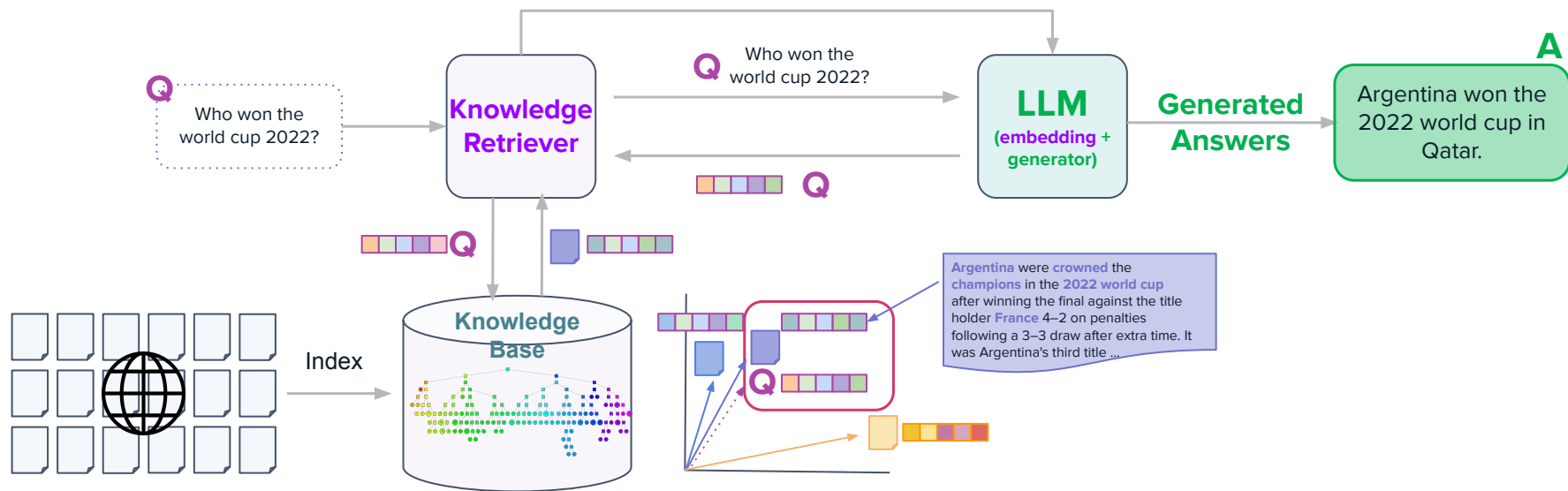
RAG



RAG Workflow

RAG

Q [] + []





RAG Workflow

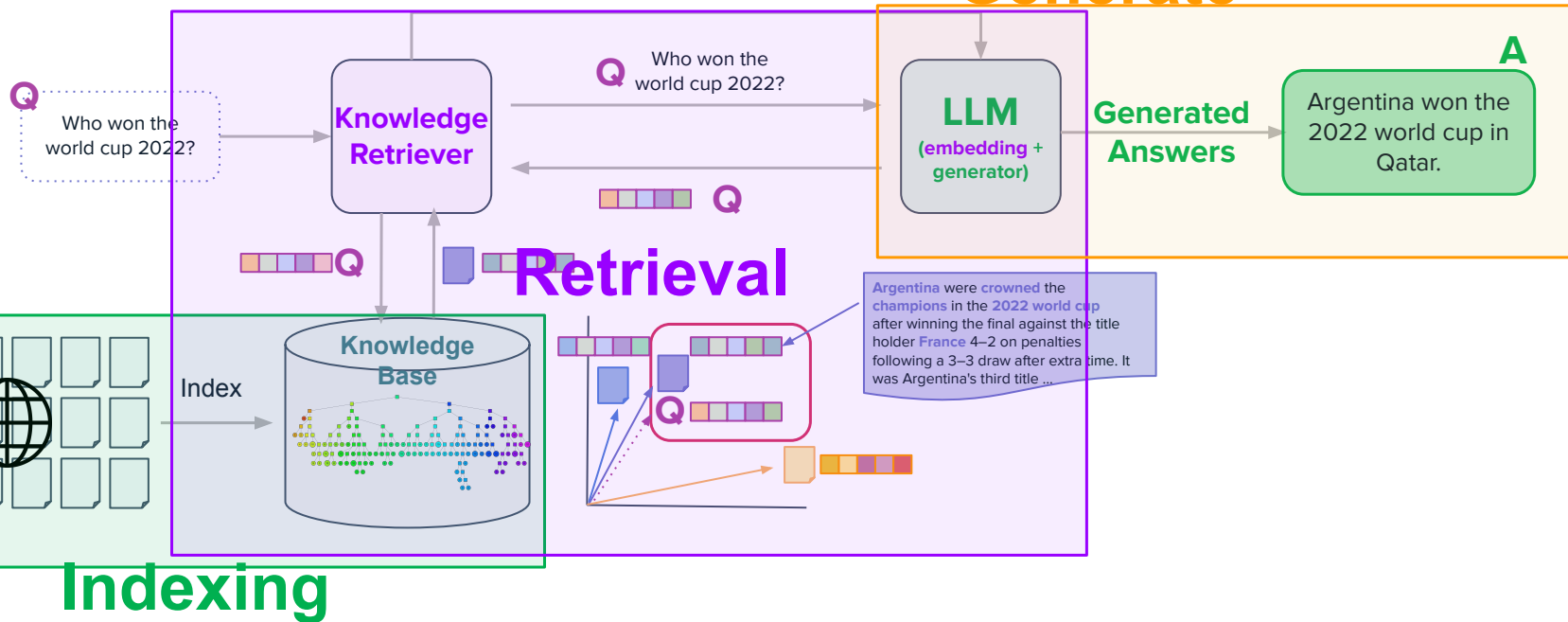
RAG

The Three stages of RAG.

Query + Context
(retrieved)



Augment &
Generate





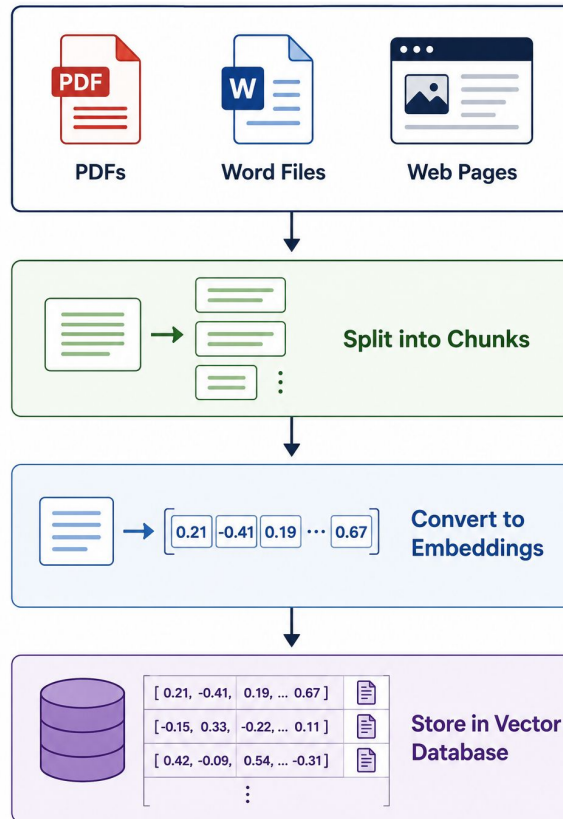
What is Indexing?

RAG

Indexing is the process of preparing your documents so they can be searched efficiently.

- Instead of searching entire PDFs every time,

RAG prepares them in advance.





Why and When Indexing is used?

RAG

Why?

- Makes searching much faster
- Enables semantic search
- Prepared in advance
(Updated when documents or the indexing strategy change.)

When?

- Before users start asking questions
- Whenever new documents are added



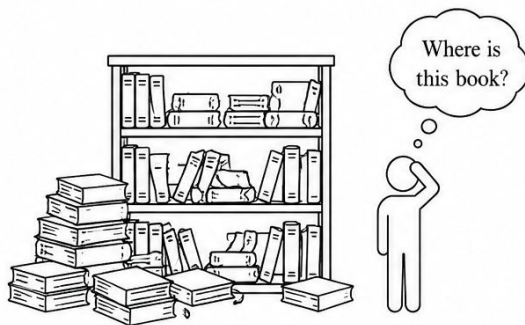
Indexing Analogy

RAG

Imagine organizing a library before anyone visits.

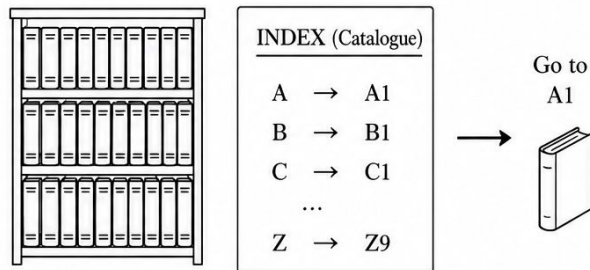
Without organizing

finding books becomes slow.



With organizing (Indexing)

finding books becomes fast.





Preparing documents for Search

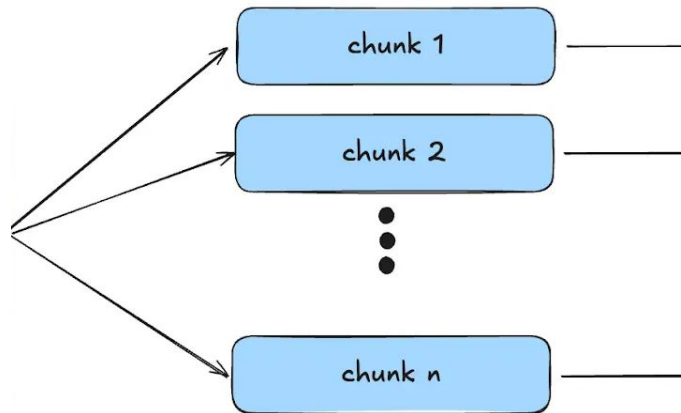
RAG

Before documents can be indexed...

Simply storing an entire document is **not enough**.

During the indexing process, documents are first **prepared** so that the system can search them efficiently.

One important preparation step is called **chunking**.



<https://www.elastic.co/search-labs/blog/chunking-strategies-elasticsearch>

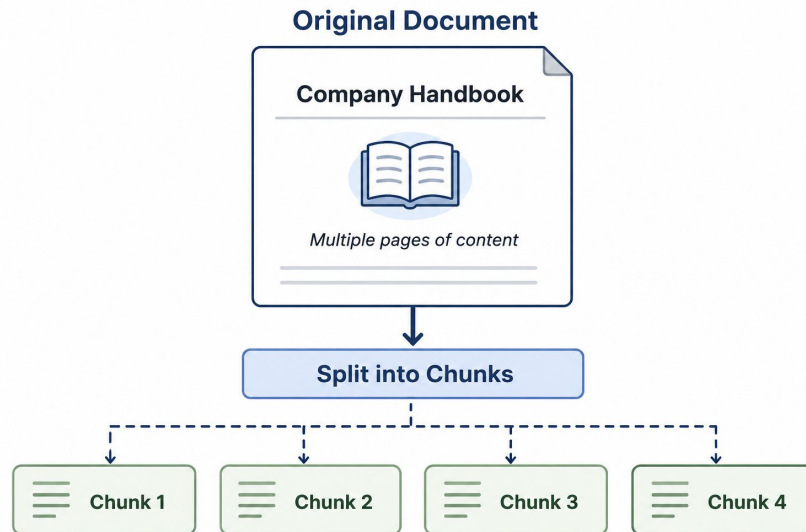


Preparing documents for Search

RAG

Chunking is the process of dividing a large document into **smaller, meaningful sections**, called **chunks**.

Each chunk represents a portion of the document that can later be searched independently.





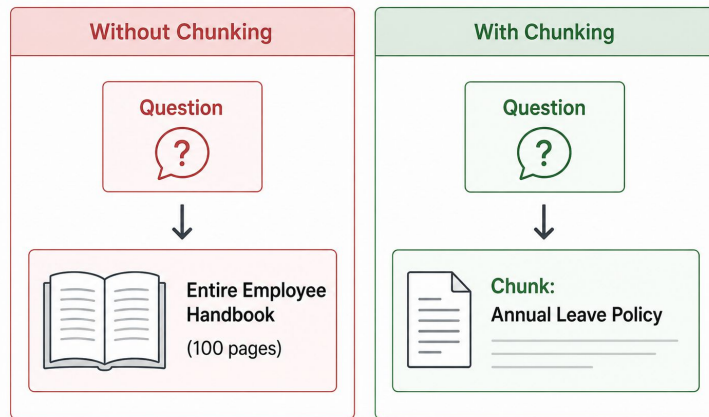
Why do we need Chunking?

RAG

Chunking is the process of dividing a large document into **smaller, meaningful sections**, called **chunks**.

Each chunk represents a portion of the document that can later be searched independently.

Example

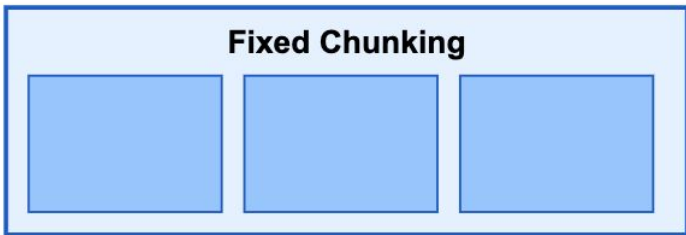




Different Chunking Strategies

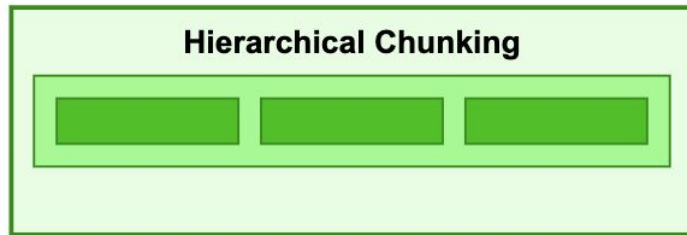
RAG

Fixed Chunking



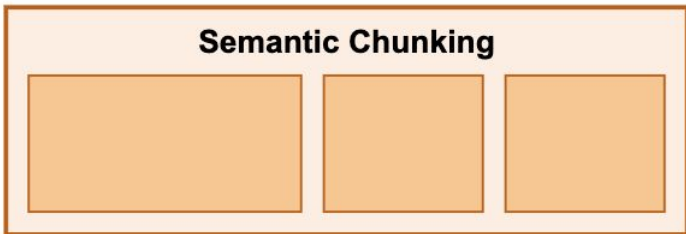
Fixed-size chunks with optional overlap

Hierarchical Chunking



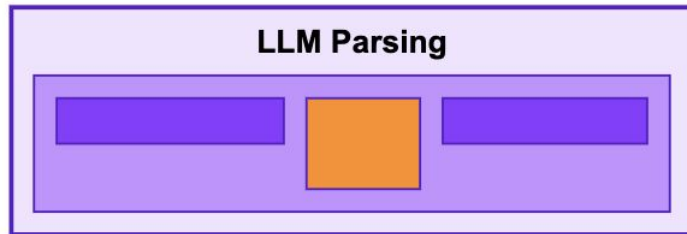
Parent and child chunks

Semantic Chunking



Chunks based on semantic similarity

LLM Parsing



Custom parsing with LLM instructions



Benefits of Chunking

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Faster retrieval

Reduced hallucinations

More relevant context

Better answer quality

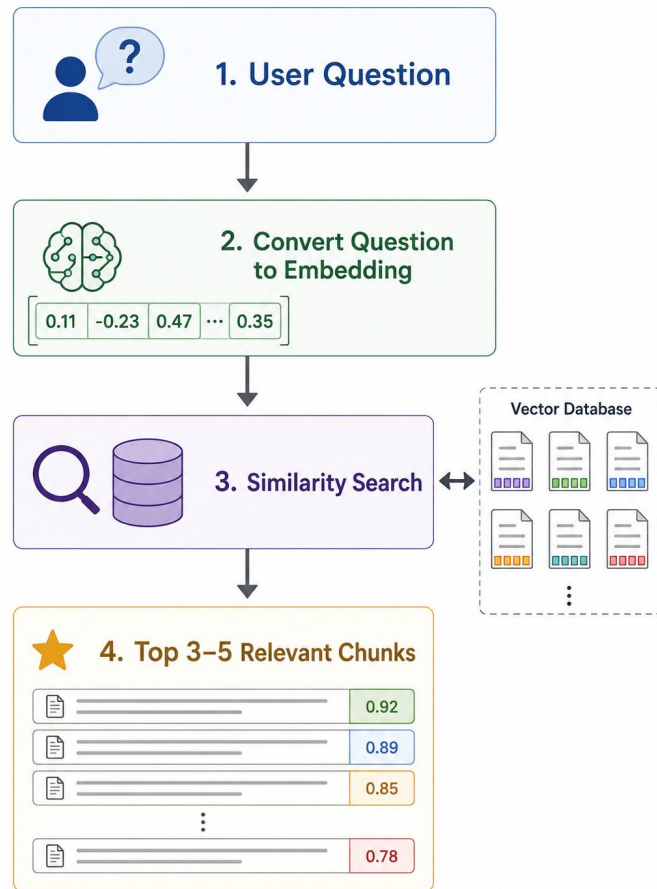


What is Retrieval?

RAG

Retrieval finds the most relevant document chunks for a user's question.

- Instead of reading every document, the system searches for **the most similar pieces of information.**





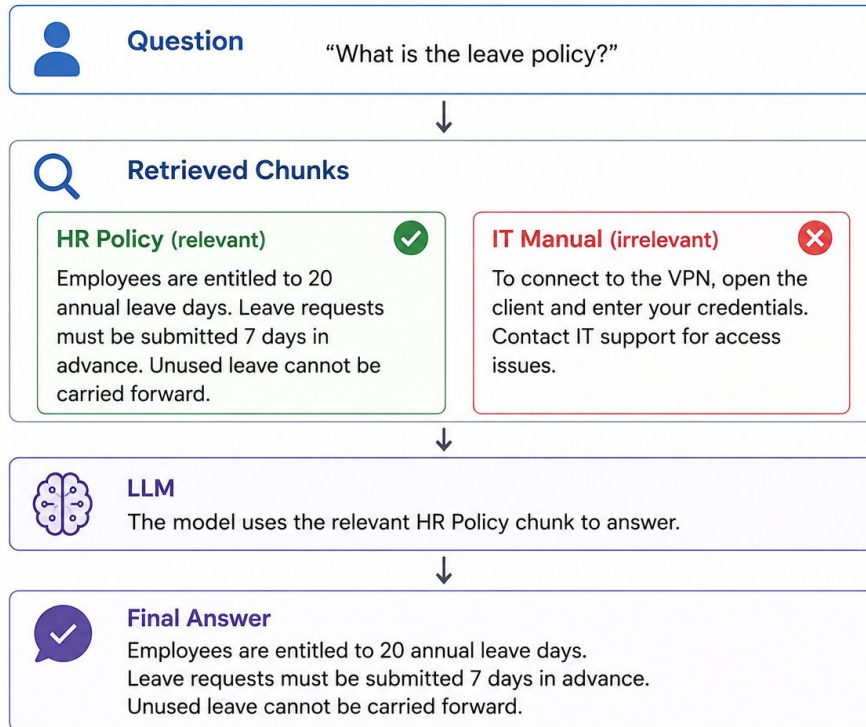
Why do we Retrieve?

RAG

Why?

- Faster than searching every document
- Provides relevant context
- Reduces hallucinations

Example:



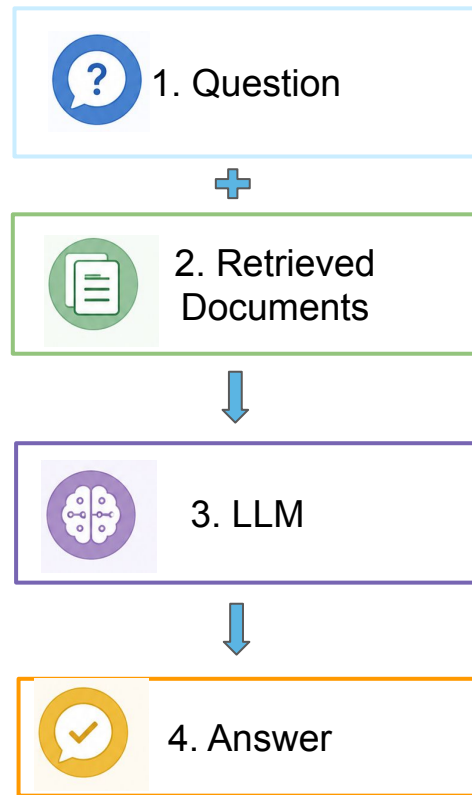


What is Augmentation?

RAG

The retrieved documents are added to the user's prompt. The LLM now answers using both

- the user's question
- the retrieved context





Augmentation Example

RAG



1. Question

What is the company's leave policy?



2. Retrieved Documents

- Employees receive 20 annual leave days.
- Leave requests must be submitted 7 days in advance.
- Unused leave cannot be carried forward.



3. LLM

The model reads the question and the retrieved documents to generate an answer.



4. Grounded Answer

Employees are entitled to 20 annual leave days. Leave requests must be submitted 7 days in advance, and unused leave cannot be carried forward.



Vector DB RAG

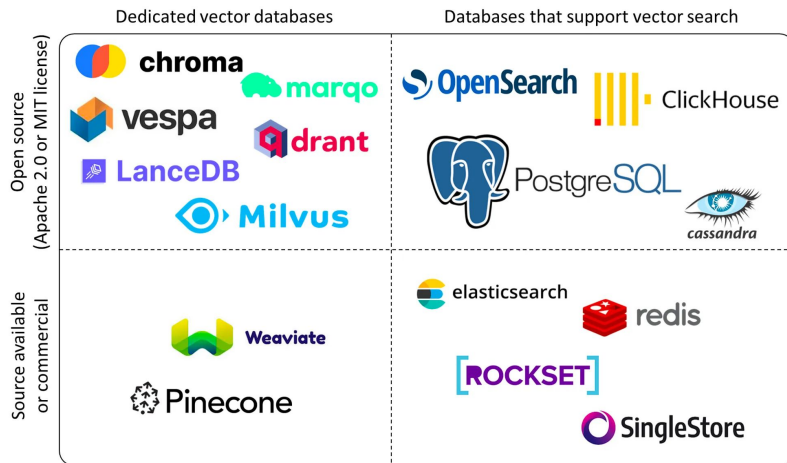


What is Vector DB?

RAG

A vector database stores embeddings together with IDs, metadata, and often the original text chunks.

Rather than searching using exact keywords, it searches using
meaning (semantic similarity).

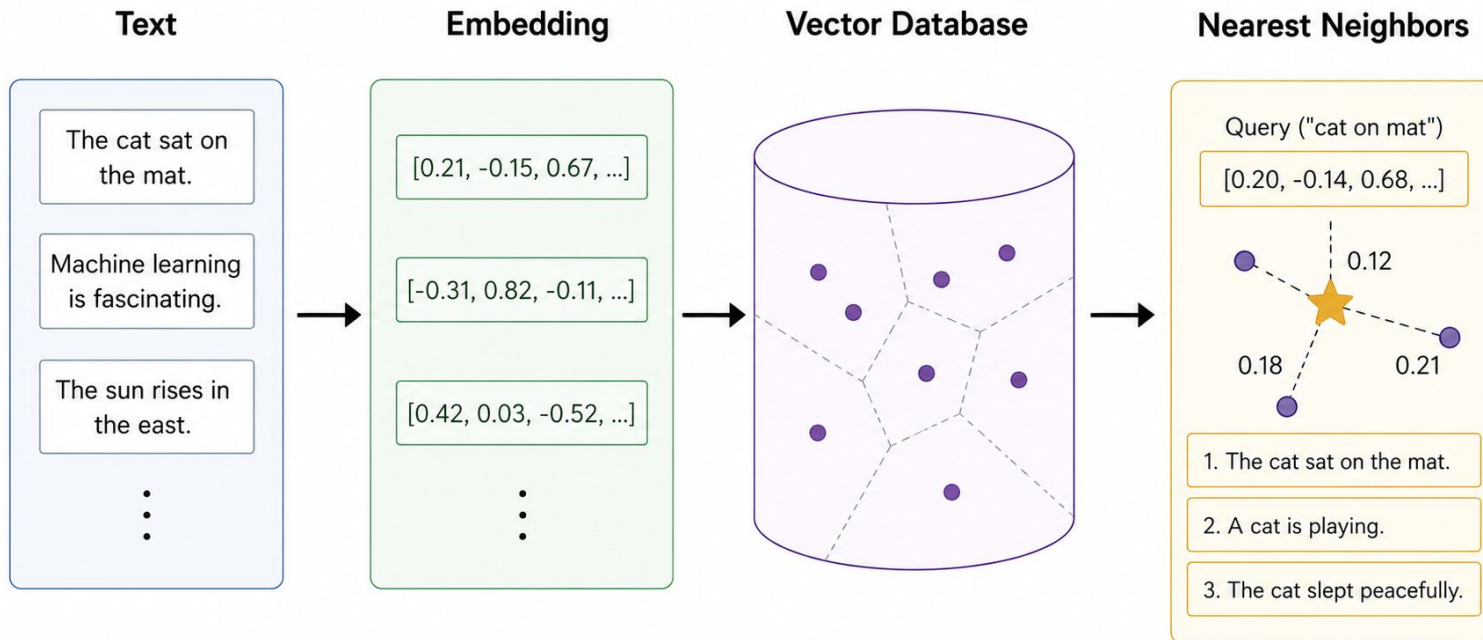


<https://www.datacamp.com/blog/the-top-5-vector-databases>



What is Vector DB?

RAG





Why and When to use VectorDB?

RAG

Why?

- Finds information by meaning, not only exact words
- Retrieves relevant chunks from large document collections
- Supports fast similarity search over embeddings

When?

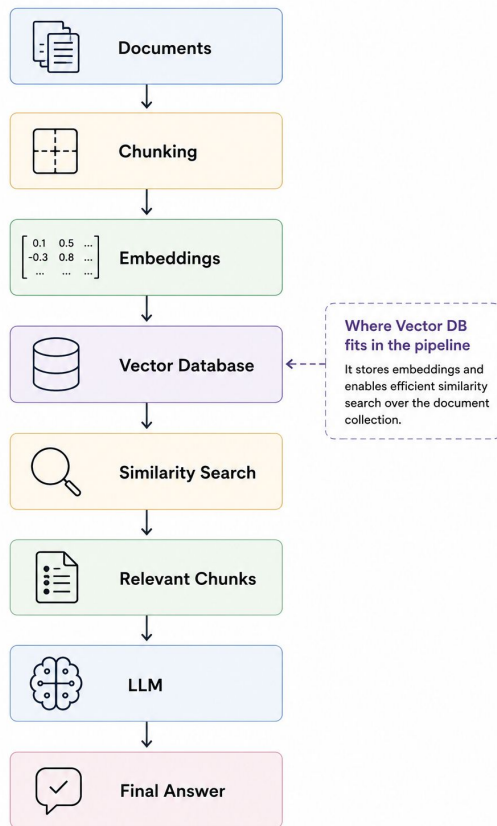
Use a vector database when you need:

- Semantic search
- RAG over documents
- Recommendation systems
- Similarity-based image or text search



Where Vector DB Fits in the Pipeline

RAG





Difference between traditional DB and Vector DB

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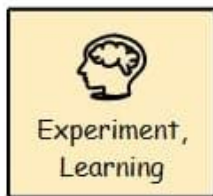
Traditional DB	Vector DB
Searches exact keywords	Searches semantic meaning
SQL queries	Similarity Search
Good for structured data	Good for Embeddings
Rows & Columns	High Dimensional Vectors

Vector search is often combined with keyword search and metadata filters.



Popular Vector DB

RAG



FAISS



Pinecone, Weaviate



Qdrant, Weaviate



Milvus



devopscube.com



Where Should You Start?

RAG

Your Goal	A Good Starting Point
Learn RAG locally	Chroma or FAISS
Build a simple prototype	Chroma or Qdrant
Self-host a production system	Qdrant, Milvus, or Weaviate
Use a managed service	Pinecone or a cloud-managed option



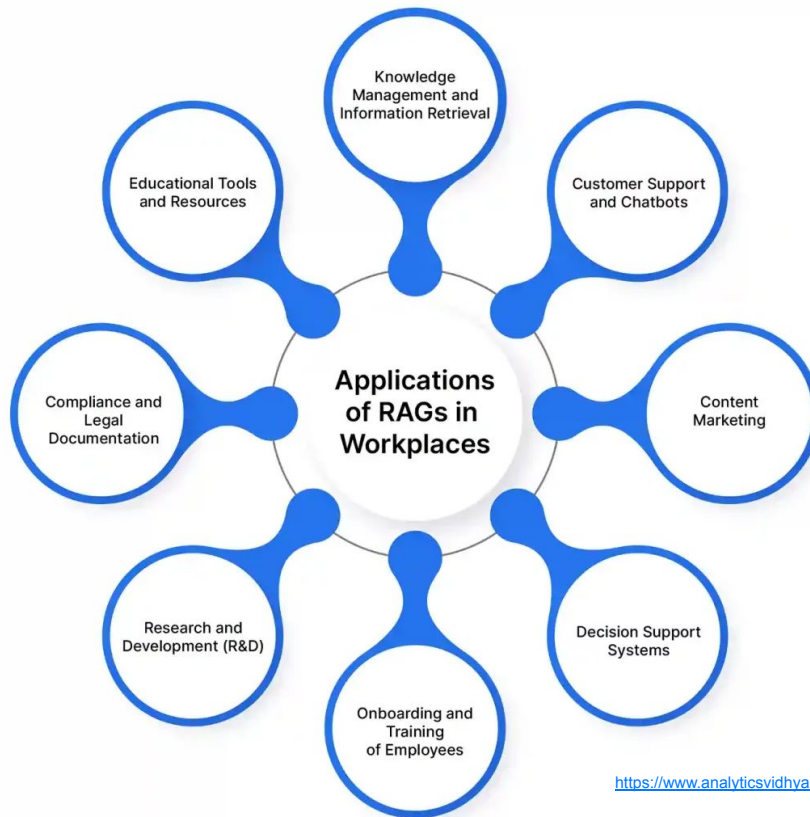
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Real World Applications RAG



Real World Applications

RAG



<https://www.analyticsvidhya.com/blog/2024/10/applications-of-rag-in-workplaces/>



RAG Vs Fine tuning

RAG

RAG	Fine-Tuning
Adds external knowledge	Changes model behavior
No Retraining	Requires additional training
Easy to Update	Expensive to update
Best for changing information	Best for learning new Tasks



Summary

- Understood why Large Language Models alone are insufficient for answering questions about private or up-to-date information.
- Learned what Retrieval-Augmented Generation (RAG) is and when it should be used.
- Explored the three key stages of a RAG pipeline: **Indexing**, **Retrieval**, and **Augment & Generate**.
- Learned how documents are prepared through chunking and embeddings before being indexed for efficient retrieval.
- Understood how semantic retrieval finds the most relevant information using embeddings rather than exact keyword matching.
- Explored the role of **Vector Databases** and how they store and search embeddings efficiently.
- Compared popular vector databases such as **FAISS**, **Chroma**, **Milvus**, **Weaviate**, **Pinecone**, and **Qdrant**, along with their common use cases.
- Identified the benefits and limitations of RAG and understood how it differs from fine-tuning.
- Explored real-world applications where RAG powers intelligent, up-to-date AI assistants and enterprise search systems.



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!